

XINYAN LI 李心妍

LLM Post-Training | Tool-Integrated Reasoning

SFT + RL • Mathematical Reasoning • Efficient Inference

📍 Hong Kong / Open to relocation

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🎓 Google Scholar 🌐 Website

PROFILE LLM research engineer with hands-on experience post-training 30B+ models through SFT and reinforcement learning. Current work focuses on tool-integrated mathematical reasoning and token-efficient inference; doctoral research focused on privacy leakage in encrypted search systems.

🧰 EXPERIENCE

Senior Research Engineer

Jun 2025–Present

HUAWEI HONG KONG RESEARCH CENTER • HONG KONG

- Independently lead end-to-end post-training of **30B+ LLMs**, from data processing and SFT/RL through distributed rollout and evaluation, with a focus on tool-integrated and token-efficient reasoning.

Research Intern

Dec 2024–May 2025

HUAWEI HONG KONG RESEARCH CENTER • HONG KONG

- Explored RAG architectures and input-attribution methods to understand how individual input tokens affect model outputs; analyzed business failure cases and evaluated retrieval-based mitigations.

Research Assistant

Dec 2020–Jun 2021

CITY UNIVERSITY OF HONG KONG • HONG KONG

Intern Student

Jun 2018–Jul 2018

CHINA TELECOM • INTERNET INNOVATION CENTER

🎓 EDUCATION

Ph.D. in Computer Science

Sep 2021–Apr 2025

CITY UNIVERSITY OF HONG KONG

GPA: 3.87/4.0 | Thesis: *Designing Leakage-Aware Mechanisms for Encrypted Search Systems*

Supervisor: Prof. Cong Wang —IEEE Fellow, Head of Computer Science & Chair Professor of Cyber Security

Institutional Research Tuition Scholarship, 2022–2024

M.Sc. in Computer Science, with Distinction

Sep 2019–Jun 2020

CITY UNIVERSITY OF HONG KONG

Department of Computer Science Achievement Scholarship (top three students), 2019

B.Eng. in Computer Science and Technology

Sep 2015–Jun 2019

SOUTH CHINA UNIVERSITY OF TECHNOLOGY

Academic exchanges: Tianjin University, Sep 2017–Feb 2018; ITMO University, Russia, Sep–Oct 2018

🏆 SELECTED RESEARCH & PROJECTS

Tool-Integrated Mathematical Reasoning

- Trained LLMs with SFT and multi-stage RL to use Python and Wolfram/Mathematica for mathematical problem solving, targeting more reliable answers through executable computation.
- Developed capability-specific training and evaluation across K–12 mathematics and physics, undergraduate mathematics and physics, and competition mathematics.
- Addressed core data and optimization challenges: constructing teacher trajectories, filtering invalid or ineffective tool calls, cleaning SFT data, and designing multi-stage RL that improves genuine tool mastery.

Token-Efficient Reasoning via Long-to-Short RL

- Applied Long-to-Short RL to transfer reasoning capability from a model operating with up to a **32K context budget** into an **8K-budget model** while preserving answer accuracy.
- Designed the RL algorithm and length penalties to balance compression, accuracy, and premature termination.

 SELECTED RESEARCH & PROJECTS (CONTINUED)

Agentic LLM Training — Technical Survey

- Studied agent training through SFT, multi-turn RL, tool-use trajectories, environment feedback, verifiable rewards, and trajectory curation to understand agentic training pipelines and engineering bottlenecks.

 SELECTED PUBLICATIONS

- X. Li, Y. Du, and C. Wang. “When Differential Privacy Meets Query Control: A Hybrid Framework for Practical Range Query Leakage Quantification and Mitigation”. *IEEE TSC*, 18(2):1137–1151, 2025. [\[DOI\]](#)
- X. Li, Y. Du, and C. Wang. “RangeQC: A Query Control Framework for Range Query Leakage Quantification and Mitigation”. *IEEE ICDCS*, 749–759, 2023. [\[DOI\]](#)
- X. Li, Y. Chen, C. Wang, and C. Shen. “When Deep Learning Meets Differential Privacy: Privacy, Security, and More”. *IEEE Network*, 35(6):148–155, 2021. [\[DOI\]](#)

 TECHNICAL SKILLS

POST-TRAINING	SFT, reinforcement learning, multi-stage RL, Long-to-Short RL
REASONING	Tool-integrated reasoning, tool calling, mathematical and physics reasoning
FRAMEWORKS	verl, TRL
SYSTEMS	Slurm, distributed training, rollout, and evaluation workflows
PROGRAMMING	Python (primary); working knowledge of C/C++, Java, Shell, and SQL

 INTERESTS

-  Badminton  Fitness  Jigsaw puzzles  Pets